

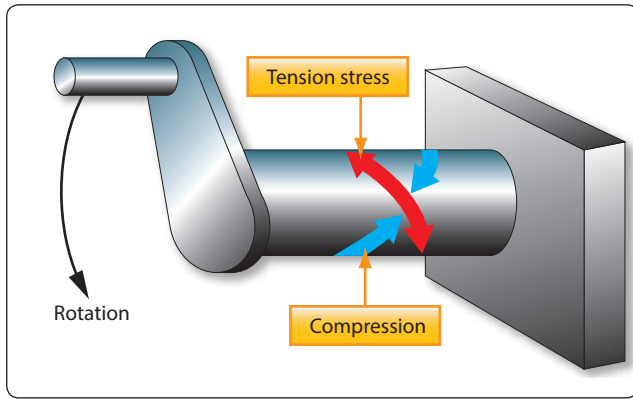


Figure 5-20. Torsion on a rotating shaft, made up of tension and compression.

length from the blades to the center of the shaft, and creates the torque that causes rotation. [Figure 5-21]

Bending

An airplane in flight experiences a bending force on the wing as aerodynamic lift tries to raise the wing. This force of lift causes the skin on the top of the wing to compress and the skin on the bottom of the wing to be under tension. When the airplane is on the ground sitting on its landing gear, the force of gravity tries to bend the wing downward, subjecting the bottom of the wing to compression and the top of the wing to tension. [Figure 5-22] During the testing that occurs prior to FAA certification, an airplane manufacturer intentionally bends the wing up and down to make sure it can take the stress without failing.

Shear

When a shear stress is applied to an object, the force tries to cut or slice through, like a knife cutting through butter. A clevis bolt, which is often used to secure a cable to a part

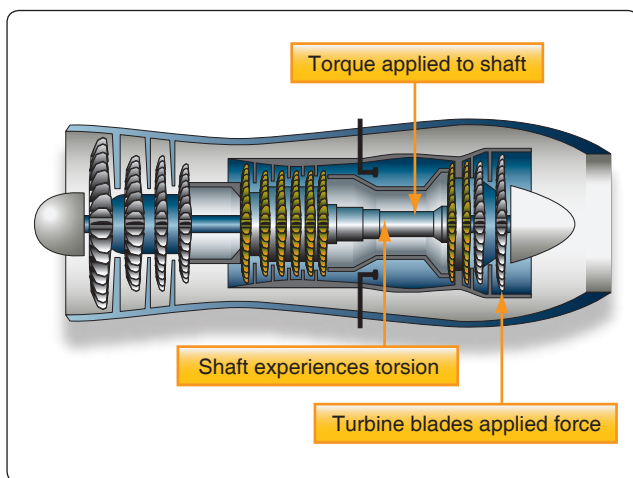


Figure 5-21. Turbofan engine, torque creating torsion in the shaft.

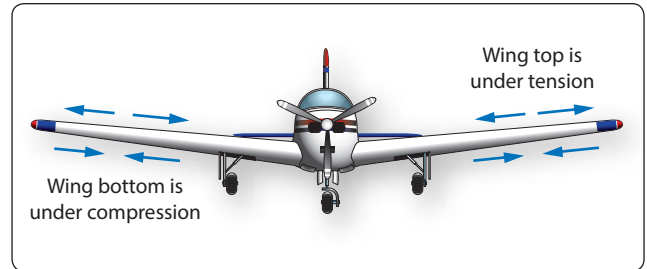


Figure 5-22. Airplane on the ground, wing under tension and compression.

of the airframe, has shear stress acting on it. As shown in Figure 5-23, a fork fitting is secured to the end of the cable, and the fork attaches to an eye on the airframe with the clevis bolt. When the cable is put under tension, the fork tries to slide off the eye by cutting through the clevis bolt. This bolt would be designed to take very high shear loads.

Strain

If the stress acting on an object is great enough, it can cause the object to change its shape or to become distorted. One characteristic of matter is that it tends to be elastic, meaning it can be forced out of shape when a force is applied and then return to its original shape when the force is removed. When an object becomes distorted by an applied force, the object is said to be strained.

On turbine engine test cells, the thrust of the engine is typically measured by what are called strain gages. When the force, or thrust, of the engine is pulling out against the strain gages, the amount of distortion is measured and then translated into the appropriate thrust reading.

A deflecting beam style of torque wrench uses the strain on the drive end of the wrench and the resulting distortion of the beam to indicate the amount of torque on a bolt or nut. [Figure 5-24]

Motion

The study of the relationship between the motion of bodies or objects and the forces acting on them is often called the study of “force and motion.” In a more specific sense, the

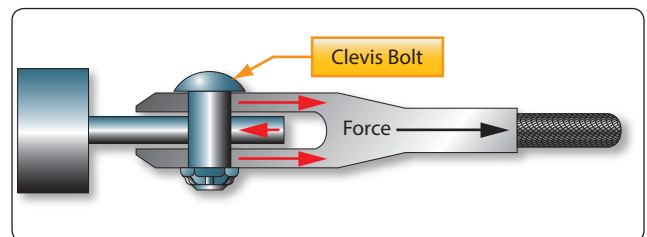


Figure 5-23. Clevis bolt, red arrows show opposing forces trying to shear the bolt.